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B.Tech. Degree V Semester Regular/Supplementary Examination January 2023

ME 19-205-0502 MECHANICS OF MACHINERY (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Understand the principles of kinematic pairs, chains and their classification, DOF, inversions, equivalent chains and planar mechanisms.
- CO2: Analyze the planar mechanisms for position, velocity, and acceleration.
- CO3: Synthesize planar four bar and slider crank mechanisms for specified kinematic conditions.
- CO4: Evaluate gear tooth geometry and select appropriate gears for the required applications.
- CO5: Design cams and followers for specified motion profiles.
- Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
- PO – Programme Outcome

PART A

(Answer **ALL** questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Define straight line mechanisms. Describe one type of exact straight line motion mechanism with the help of a sketch.	3	3	L1	1	1
(b) Locate all the instantaneous centres for a four bar mechanism. The lengths of various links are: AD = 125 mm; AB = 62.5 mm; BC = CD = 75 mm; ∠BAD = 60° If the link AB rotates at a uniform speed of 10 r.p.m. in the clockwise direction, find the angular velocity of the links BC and CD.	3	3	L3	2	2
(c) Derive an expression for the length of the arc of contact in a pair of meshed spur gears.	3	3	L2	4	1
(d) Explain reverted gear train with a neat sketch.	3	3	L1	4	2
(e) Discuss the displacement, velocity and acceleration diagrams for a follower when it moves with simple harmonic motion.	3	3	L2	5	1
(f) A conical pivot bearing supports a vertical shaft of 200 mm diameter. It is subjected to a load of 30 kN. The angle of the cone is 120° and the coefficient of friction is 0.025. Find the power lost in friction when the speed is 140 r.p.m., assuming (i) uniform pressure (ii) uniform wear.	3	3	L3	5	2
(g) Derive the formula for the maximum torque transmitted by a single plate clutch of external and internal radii r_1 and r_2 , if the limiting coefficient of friction is μ and the axial spring load is W . Assume that the pressure intensity on the contact faces is uniform.	3	3	L3	3	3
(h) Derive the Freudenstein's equation for the synthesis of a four bar mechanism.	3	3	L1	3	2

(P.T.O.)

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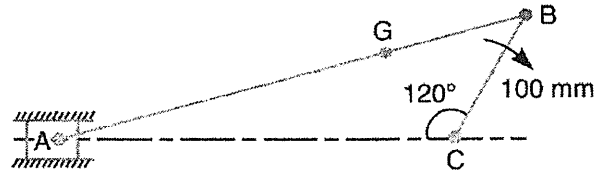
PART B

(4 × 12 = 48)

- | | Marks | BL | CO | PO |
|---|-------|----|----|----|
| II. (a) A crank and slotted lever mechanism used in a shaper has a centre distance of 300 mm between the centre of oscillation of the slotted lever and the centre of rotation of the crank. The radius of the crank is 120 mm. Find the ratio of the time of cutting to the time of return stroke. | 6 | L3 | 1 | 2 |
| (b) Identify the kinematic chains to which the following mechanisms belong and explain the mechanism with a neat sketch
(i) Whitworth quick return motion mechanism
(ii) Elliptical trammels. | 6 | L2 | 1 | 2 |

OR

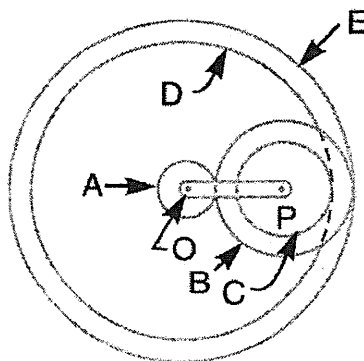
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|---|----|----|---|---|
| III. (a) An engine mechanism is shown in the following figure. The crank $CB = 100$ mm and the connecting rod $BA = 300$ mm with centre of gravity G , 100 mm from B . In the position shown, the crankshaft has a speed of 75 rad/s and an angular acceleration of 1200 rad/s^2 . Find:
(i) Velocity of G and angular velocity of AB
(ii) Acceleration of G and angular acceleration of AB . | 12 | L3 | 2 | 2 |
|---|----|----|---|---|



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|--|---|----|---|---|
| IV. (a) Two gears of module 4 mm have 24 and 33 teeth. The pressure angle is 20° and each gear has a standard addendum of one module. Find the length of arc of contact and the maximum velocity of sliding if the pinion rotates at 120 r.p.m. | 4 | L3 | 4 | 2 |
| (b) Two gear wheels mesh externally and are to give a velocity ratio of 3. The teeth are of involute form of module 6 mm. The standard addendum is 1 module. If the pressure angle is 18° and pinion rotates at 90 r.p.m., find : (i) the number of teeth on each wheel, so that the interference is just avoided (ii) the length of the path of contact (iii) the maximum velocity of sliding between the teeth. | 8 | L3 | 4 | 2 |

OR

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|---|----|----|---|---|
| V. (a) The following figure shows diagrammatically a compound epicyclic gear train. Wheels A, D and E are free to rotate independently on spindle O, while B and C are compound and rotate together on spindle P, on the end of arm OP. All the teeth on different wheels have the same module. A has 12 teeth, B has 30 teeth and C has 14 teeth cut externally. Find the number of teeth on wheels D and E which are cut internally. If the wheel A is driven clockwise at 1 r.p.s. while D is driven counter clockwise at 5 r.p.s., determine the magnitude and direction of the angular velocities of arm OP and wheel E. | 12 | L3 | 4 | 2 |
|---|----|----|---|---|



(Continued)

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|-----------|---|----|----|---|---|
| VI. | Design a cam for operating the exhaust valve of an oil engine. It is required to give equal uniform acceleration and retardation during opening and closing of the valve each of which corresponds to 60° of cam rotation. The valve must remain in the fully open position for 20° of cam rotation. The lift of the valve is 37.5 mm and the least radius of the cam is 40 mm. The follower is provided with a roller of radius 20 mm and its line of stroke passes through the axis of the cam. | 12 | L3 | 5 | 2 |
| OR | | | | | |
| VII. (a) | A body, resting on a rough horizontal plane required a pull of 180 N inclined at 30° to the plane just to move it. It was found that a push of 220 N inclined at 30° to the plane just moved the body. Determine the weight of the body and the coefficient of friction. | 4 | L3 | 5 | 2 |
| (b) | The pitch of 50 mm mean diameter threaded screw of a screw jack is 12.5 mm. The coefficient of friction between the screw and the nut is 0.13. Determine the torque required on the screw to raise a load of 25 kN, assuming the load to rotate with the screw. Determine the ratio of the torque required to raise the load to the torque required to lower the load and also the efficiency of the machine. | 8 | L3 | 5 | 2 |
| VIII. (a) | A plate clutch has three discs on the driving shaft and two discs on the driven shaft, providing four pairs of contact surfaces. The outside diameter of the contact surfaces is 240 mm and inside diameter 120 mm. Assuming uniform pressure and $\mu = 0.3$; find the total spring load pressing the plates together to transmit 25 kW at 1575 r.p.m. If there are 6 springs each of stiffness 13 kN/m and each of the contact surfaces has worn away by 1.25 mm, find the maximum power that can be transmitted, assuming uniform wear. | 8 | L3 | 3 | 2 |
| (b) | In a laboratory experiment, the following data were recorded with rope brake:
Diameter of the flywheel = 1.2 m; Diameter of the rope = 12.5 mm;
Speed of the engine = 200 r.p.m.; Dead load on the brake = 600 N;
Spring balance reading = 150 N.
Calculate the brake power of the engine. | 4 | L3 | 3 | 2 |
| OR | | | | | |
| IX. | Synthesize a four-bar mechanism to generate a function $y = \sin x$ for $0 \leq x \leq 90^\circ$. The range of the output crank may be chosen as 60° while that of input crank be 120° . Assume three precision points which are to be obtained from Chebyshev spacing. Assume fixed link to be 52.5 mm long and $\theta_1 = 105^\circ$ and $\phi_1 = 66^\circ$. | 12 | L3 | 3 | 2 |

Bloom's Taxonomy Levels

L1 = 15%, L2 = 15%, L3 = 70%.

